SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. TEAM AND PROJECT OVERVIEW

Team Name

sticks and stones

Project / Solution Name

Resolve

Name (Leader)

Ee Jun Hao

Email & Phone (Leader)

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University Affiliation(s)

XIAMEN UNIVERSITY MALAYSIA

Live Prototype URL

https://resolve-sdc-hackathon-src.vercel.app/

Code Repository URL (public / access granted to judges)

gh repo clone ejhchesteree7-code/Resolve-SDC-HACKATHON-

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Resolve is a web app hosted on Vercel. Flow:

Interface: a React + Vite + Tailwind app. Guests use a simulated public review page (standing in for Google); staff use the case dashboard with a simulated role switcher (Staff, Manager, Owner).

Backend: an Express API running as a Vercel serverless function. Every action passes server-side role, tenant and outlet authorization.

AI step: the API sends review text to an optional OpenAI-compatible model for classification and a draft reply, with strict JSON validation and an 8-second timeout. If AI is off or fails, a built-in keyword fallback (Basic mode) takes over, so the workflow never depends on AI.

Rules: plain routing rules (R1 to R6, voucher limit) assign the case to Staff, Manager or Owner. Nothing is posted until a human approves.

Database: Supabase Postgres stores reviews, users and tenants, plus an append-only audit log enforced by a database trigger.

Simulated: Google Business Profile and the private WhatsApp follow-up. No message leaves the system.

Development: Replit Agent.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React 19.1 + Vite 7	Frontend UI and build	MIT
Tailwind CSS 4	Styling	MIT
Express 5	Backend API (Vercel function)	MIT
Drizzle ORM 0.45	Typed database queries	Apache-2.0
Zod 3	Validates requests and AI output	MIT
OpenAI SDK 7	Optional AI triage (off in demo)	Apache-2.0

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI-compatible chat API	Optional review classification and reply drafting	Paid; switched off in the live demo (Basic mode)
Supabase Postgres	Stores reviews, users and the audit log	Free tier
Google Business Profile, WhatsApp Business	Not called. Both are simulated in the demo	N/A

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
vercel (hosting)	Hosts the frontend and the API as a serverless function; deploys from GitHub	hobby (free)
supabase (backend)	Postgres database for reviews, users, audit log	free/Southeast Asia (Singapore)
Github	Public code repository; triggers Vercel deploys	N/A
Replit	Development environment and backup deployment	paid by sponsor of this hackathon

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-4.1-nano	OpenAI API	Pre-trained,Off in the live demo	OpenAI API terms of use
Keyword and template fallback (Basic mode)	Built by our team	Rule-based, not a trained model. Used when AI is off or fails	Original work (Apache-2.0 with the repo)

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Seed reviews (demo cases)	Written by our team	Synthetic	Apache-2.0 (with the repo)	N
10-review timed test set	Written by our team, in the repo	Synthetic	Apache-2.0 (with the repo)	N
Reviews typed into the demo guest page	Entered by testers at runtime	synthetic	N/A	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit Agent	Built the first version of the app: frontend, API, database schema, tests	Major
Claude (Claude Code, Sonnet 5.5)	Improved AI triage and routing, recurring-issue insight, security headers, simulated WhatsApp follow-up, tests, deployment, README and documentation	Major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We reviewed every AI-generated change before it reached the live site. Changes went to a Vercel preview, where we clicked through the full journey, then merged to main only after we approved. We wrote automated tests (permissions, routing, AI fallback, audit log), and the code runs them. We also ran a security header scan and fixed the issues it found. Product decisions, scope and the target customer were chosen by us. In the product itself, AI only drafts: a human approves every reply and voucher. We take full responsibility for the submitted work.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
SDC organisers / Alpha Red Solutions (Orbyte challenge)	Challenge brief	Mentor feedback shaped our focus on one customer type
Replit	[credits / plan provided for the hackathon, or delete this row]	Development with Replit Agent and backup hosting

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The USD 20 Replit credits helped us build faster, but the prototype does not depend on them. The app is plain React, Express and Postgres code, so it runs on any host. We already moved the live demo to Vercel and Supabase. The challenge brief and mentoring shaped our focus and design, but no sponsor API, data or credits run inside the product, so nothing needs replacing.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Resolve stores review text, rating, reviewer first name (as typed on the demo guest page), outlet, AI draft, final reply, voucher amount and an audit log of staff actions (demo user IDs only). All of it is synthetic demo data; we collect no real customer data and no contact details. Data sits in Supabase Postgres behind row-level security with no public policies; only our server connects, through a pooled connection string kept in environment variables, never in the repository. Every action is role-checked on the server, and the audit log is append-only. If an AI key is enabled, only the outlet name, rating and review text go to the AI API, not the reviewer's name. The live demo runs without AI. Demo data has no retention schedule and will be deleted after the hackathon. A real deployment would add retention limits and guest consent, especially for WhatsApp.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Limits: the AI can misclassify or draft a poor reply (hallucination), may be weaker on Manglish and mixed-language reviews, may treat vague reviews unfairly, and depends on a single provider with rate limits, cost and outages. The live demo currently runs the keyword fallback (Basic mode), which is less accurate than the AI.
Mitigations: AI output must pass strict JSON validation and an 8-second timeout, otherwise Basic mode takes over. High-severity cases (food safety, illness, billing disputes) always go to the Owner, even in Basic mode. Vague reviews are flagged as “unclear” and the reply asks for details. Nothing is posted or compensated without human approval, and every step is in an append-only audit log. Review text is treated as untrusted input in the prompt. We have not yet measured accuracy on a large real dataset.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- [ ] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [ ] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [ ] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [ ] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [ ] Any use of official SDC sponsor resources has been declared in Section 5.
- [ ] The prototype described here is accessible to judges through the links provided in Section 1.
- [ ] All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date] 23/10/2026